Analysis of the Air Quality Health Index in Hong Kong

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1 Introduction

Air pollution represents a major environmental and public health challenge in highly urbanized regions. Hong Kong regularly experiences elevated air pollution levels due to dense urban structures, intensive traffic, and regional atmospheric transport processes.

The Air Quality Health Index (AQHI), provided by the Hong Kong Environmental Protection Department, is a health risk-based air pollution indicator estimating short-term health risks associated with atmospheric pollution.

The AQHI is calculated from the combined health effects of four major pollutants: Ozone (O_3), Nitrogen Dioxide (NO_2), Sulphur Dioxide (SO_2), and particulate matter (PM_{10} / $PM_{2.5}$). The index is derived from 3-hour moving average pollutant concentrations and represents additional health risk percentages based on hospital admission statistics.

Further information about the AQHI calculation methodology is provided at: https://www.aqhi.gov.hk/en/what-is-aqhi/faqs.html

The objective of this project is to analyze the spatio-temporal distribution of AQHI values in Hong Kong and to explore temporal trends as well as spatial variability between monitoring stations. In addition, an interactive dashboard was developed to communicate the results and allow flexible exploration of the data.

The workflow integrates multi-year hourly AQHI measurements, spatial interpolation techniques, and interactive visualization tools within a reproducible analysis pipeline implemented in R.

2 Personas and User Stories

To guide the development of the information product, two personas were defined.

2.1 Persona 1 – Elementary School Teacher

The first persona represents a teacher working near a busy road in Hong Kong. The teacher is concerned about the effects of air pollution on children during outdoor activities.

User needs include:

- Understanding seasonal variations of air pollution
- Identifying safer periods for outdoor activities

2.2 Persona 2 – Data Analyst with Asthma

The second persona represents a resident with asthma who is interested in long-term developments of air pollution.

User needs include:

- Analysis of long-term AQHI trends
- Comparison of air pollution levels across districts

These personas motivated the integration of temporal trend analysis and spatial visualization components within the dashboard.

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3 Study Area

Hong Kong is a highly urbanized metropolitan region located on the southern coast of China. The territory consists of Hong Kong Island, Kowloon, and the New Territories and is characterized by complex topography and high population density.

Air quality in Hong Kong is influenced by both local emissions (e.g., traffic, industrial activity, and urban sources) and regional atmospheric transport from the Pearl River Delta. Seasonal meteorological conditions additionally affect pollutant dispersion and accumulation.

The Environmental Protection Department operates a network of monitoring stations distributed across urban, suburban, and roadside environments. This monitoring network enables spatial comparison of air pollution levels across different land-use types.

4 Data Sources

The project integrates multiple open datasets related to air quality monitoring and spatial reference data for Hong Kong.

4.1 Past AQHI Observations

Hourly AQHI measurements were obtained from the Hong Kong Environmental Protection Department.

https://www.aqhi.gov.hk/en/past-data/past-aqhi.html

Temporal resolution: Hourly Time period: 2021–2025

4.2 Monitoring Station Locations

https://www.aqhi.gov.hk/en/monitoring-network/air-quality-monitoring-stations.html

4.3 Administrative Boundaries

https://data.gov.hk/en-data/dataset/hk-had-json1-hong-kong-administrative-boundaries

4.4 Data Preparation

Preprocessing included:

- Import and merging of monthly datasets
- Handling missing values
- Harmonization of station metadata
- Conversion to spatial data using the sf package
- Creation of temporal attributes
- Aggregation to daily, monthly, and yearly levels

All preprocessing steps were implemented in R.

5 Methodology

The analysis workflow consists of three main components.

5.1 Exploratory Data Analysis

Exploratory analysis was used to evaluate temporal variability and data completeness.

5.2 Spatial Interpolation

Three interpolation approaches were applied:

Ordinary Kriging

Variogram modelling was used to account for spatial autocorrelation.

Manual Kriging

Parameters:

- Sill = 70%
- Nugget = 30%
- Range = median station distance

Inverse Distance Weighting

Power parameter = 2.

Spatial outputs were generated for monthly, yearly and hourly aggregations using 500 m resolution.

5.3 Dashboard Development

An interactive dashboard was implemented using R Shiny.

Temporal plots and spatial interpolation maps are dynamically updated through reactive filtering.

6 Information Product

The final information product is an interactive dashboard for exploring AQHI patterns across temporal and spatial dimensions.

The dashboard includes:

- Temporal trend analysis
- Hourly AQHI patterns
- Spatial interpolation maps

7 Repository

The full project repository is available at:

https://github.com/lraeuschel/AOSTD_final_assignment

The repository includes preprocessing scripts, spatial analysis workflows and the dashboard implementation.

8 Conclusion

This project demonstrates how spatio-temporal environmental datasets can be transformed into an interactive information product. The integration of temporal aggregation, spatial interpolation and interactive visualization supports intuitive exploration of AQHI patterns across Hong Kong.

Future improvements may include real-time data integration and extended statistical validation of interpolation methods.

9 Use of AI Tools

AI tools were used for language refinement and text structuring.

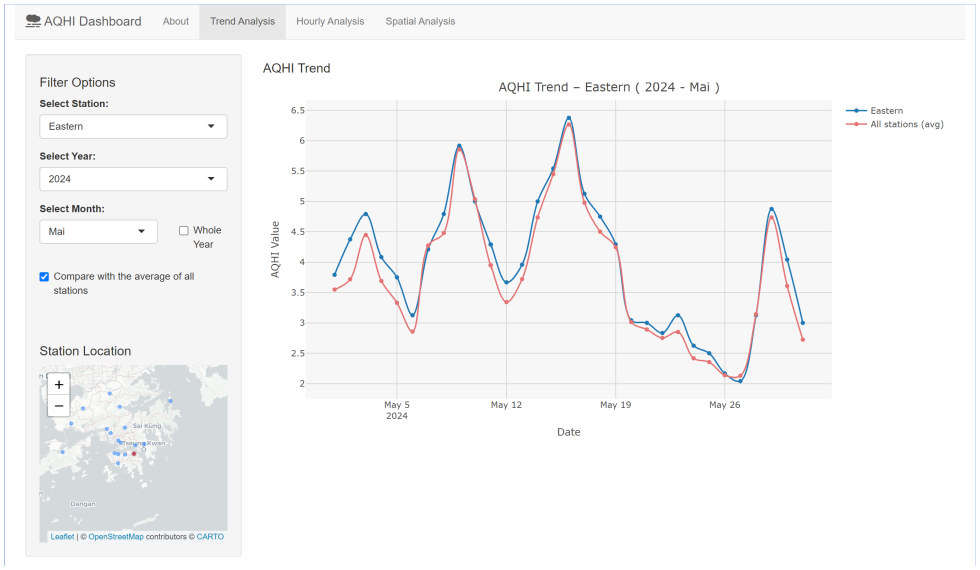

Fig. 1. Temporal AQHI trend visualization

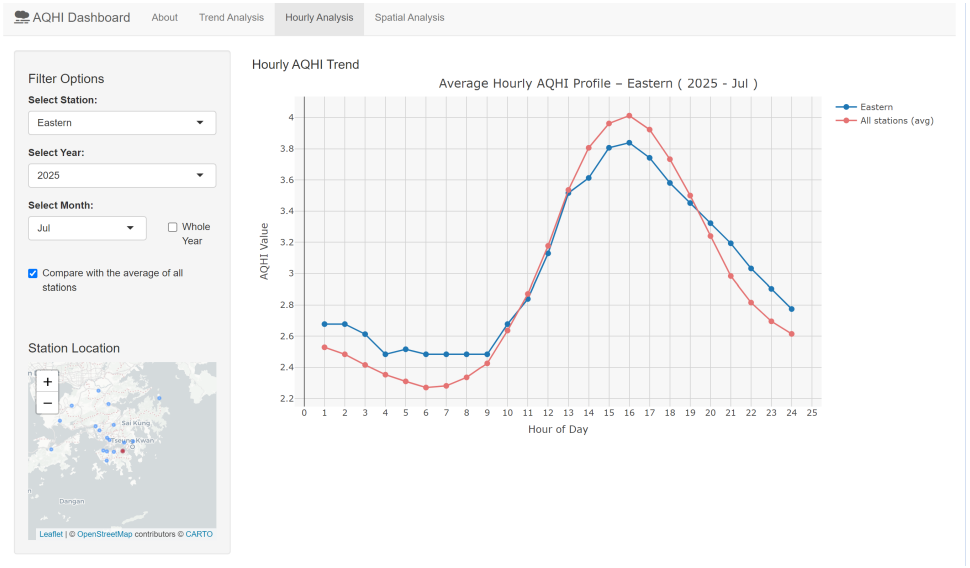

Fig. 2. Hourly AQHI patterns

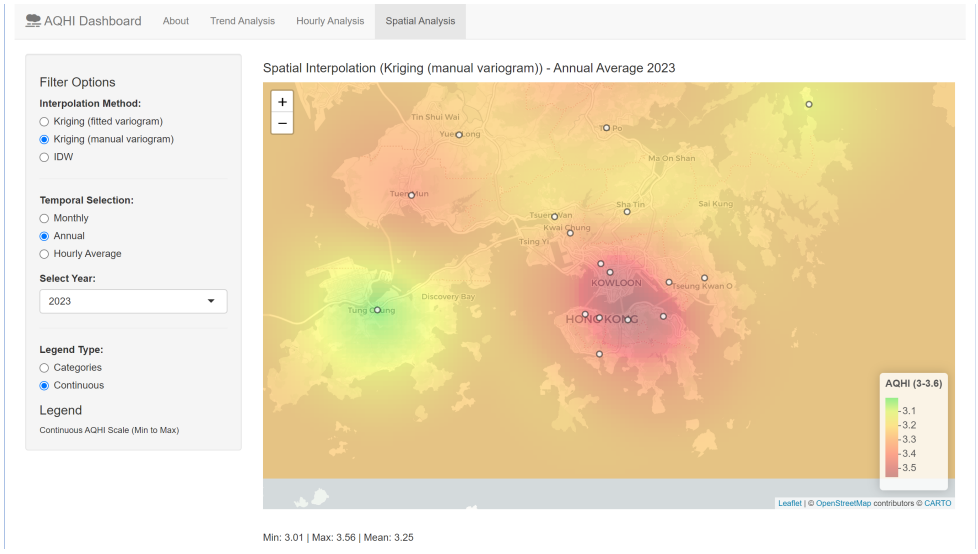

Fig. 3. Spatial interpolation results